PROGRAMME MANAGEMENT CASE STUDY

Power unit development programme control

A supplier delivery slips. Which recovery option protects the next demonstrator release, and what must management approve?

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Recommendation

Reserve the coordinated recovery option, subject to a confirmed supplier commitment and an approved EUR 31,000 quote. In this fictional case, it brings the deterministic finish forward by 8 working days and raises the illustrative on-time probability from 2.9% to 94.2%.

The committed finish boundary is 24 May 2027. The current forecast is 25 May 2027; its P80 boundary is 04 Jun 2027. Expediting supply alone meets the deterministic date but its P80 result remains beyond the commitment.

Option	Finish	P80	On time	Recovery cost
Current forecast	25 May 2027	04 Jun 2027	2.9%	EUR 0
Expedite supply	18 May 2027	25 May 2027	78.5%	EUR 18,000
Coordinated recovery	13 May 2027	19 May 2027	94.2%	EUR 31,000

Decision requested

D01: Supply chain confirms the finishing and dispatch slots. Finance validates recovery spending. The programme manager decides whether to reserve the option.
D02: The test lead protects the agreed bench sequence. Validation keeps the same test scope and acceptance criteria.
D03: The technical authority accepts release evidence separately from the forecast.

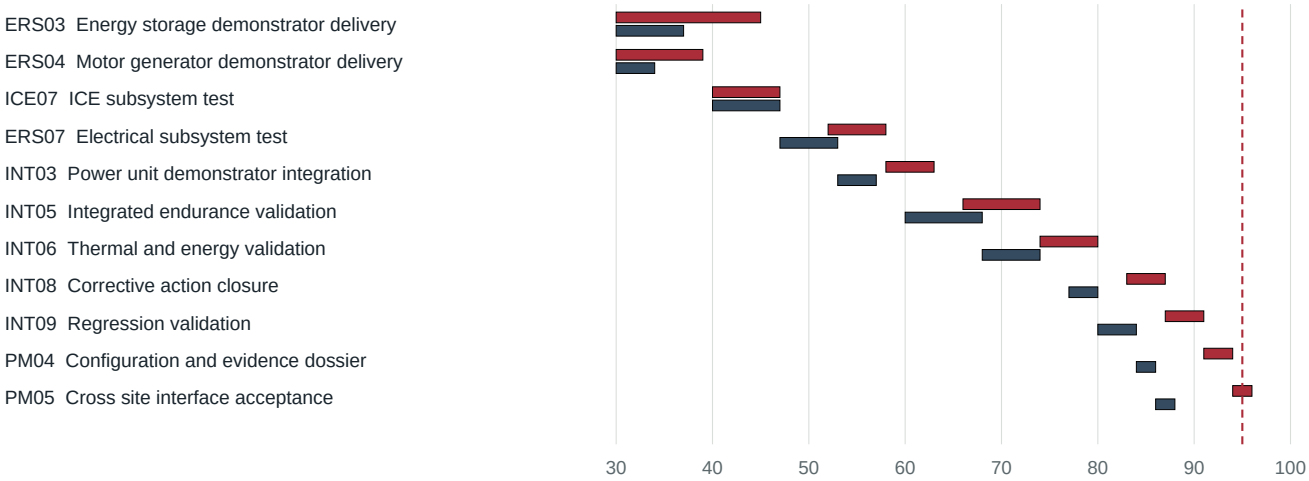
Scope

The model covers 38 activities across programme coordination, ICE, electrical systems, controls and integration. It represents a small demonstrator work package. Durations, costs, risks, resource capacity and evidence records are invented. No Audi internal data, regulatory interpretation or engineering-performance prediction is used.

SCHEDULE AND RISK

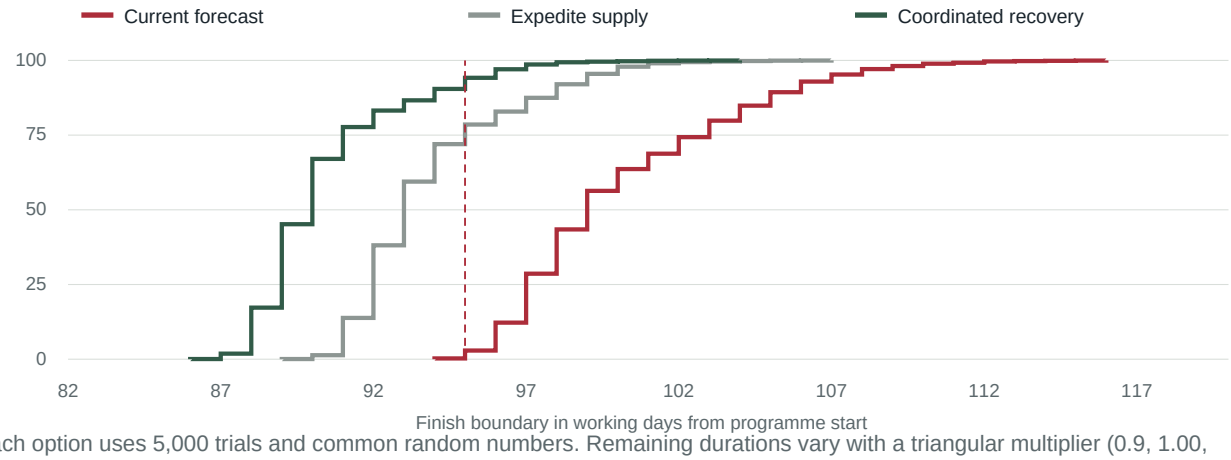
Make the consequences visible

Electrical procurement moves subsystem assembly and the shared dyno sequence. Shortening one delivery does not recover the full programme delay because a second delivery and resource reservations can become controlling constraints.



Red: current forecast. Blue: coordinated recovery. Selected activities shown; the interactive report and CSV exports include the full network.

A forecast date needs an uncertainty range



Each option uses 5,000 trials and common random numbers. Remaining durations vary with a triangular multiplier (0.9, 1.00, 1.2). One future supplier event affects both electrical deliveries; a separate event adds repeat validation. These are additional exposures, beyond the already-known delay.

Recovery assumption check	P80 finish	On-time trials
Recovery dates, but reference risk probabilities retained	21 May 2027	90.6%
Recovery with wider remaining-duration range (0.80, 1.00, 1.30)	20 May 2027	91.9%

The probabilities are illustrative planning outputs. They are sensitive to subjective durations and event assumptions. A real programme would validate these inputs with package owners, suppliers and test engineers before making a commitment.

COST AND OPERATING WORKFLOW

Keep decisions traceable

Cost at completion

EUR, fictional demonstrator scope	Current	Coordinated recovery
Actual cost at status	EUR 573,621	EUR 573,621
Owner estimate to complete	EUR 656,962	EUR 656,962
Incremental recovery	EUR 0	EUR 31,000
Late support after commitment	EUR 1,800	EUR 0
Forecast cost at completion	EUR 1,232,383	EUR 1,261,583
Baseline budget	EUR 1,185,000	EUR 1,185,000

Forecast cost reconciles to actuals plus remaining estimates, recovery spending and incremental support after the target. The model does not value an F1 championship point or infer the financial value of an earlier release. Simulated cost uncertainty includes late support only.

Weekly review workflow

1. Package owners update physical progress, actual dates, remaining duration and cost estimates.
2. The planning engine checks IDs, dependency logic, actuals and resource conflicts, then rebuilds the forecast.
3. Programme management reviews changed milestones, cost, risk and the test sequence with supply chain and finance.
4. Decisions receive an owner and due date. Evidence owners submit traceable records for technical acceptance.

Release readiness

Only 3 of 12 mandatory records are accepted at the status date. Recovery spending does not clear the other 9 records. Acceptance requires completed work, an accepted state, a reference, an approver and a date no later than the status cut-off.

Verification and practical limits

The source includes hand-calculated scheduling and earned-value tests, resource-collision tests, frozen-actual checks and evidence failure cases. Excel recalculation is checked against the Python result for all three options. The workbook keeps a fixed resource order; Python can fill new gaps when the network is rebuilt. Neither engine proves an optimal schedule. Full staffing, holiday calendars, FIA rules and other programmes' test demand need separate treatment.

Public context and references

Audi internship: Personio job 2314893. Audi power-unit context: Audi Formula Racing company overview. Scheduling reference: GAO Schedule Assessment Guide, December 2015. Accessed 6 November 2026. No formal compliance claim.